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A Facile Hydrothermal Synthesis of Three Dimensional Flower-Like NiO-Thermally Reduced Graphene Oxide (trGO) Nanocomposite for Selective Determination of Dopamine in Presence of Uric Acid and Ascorbic Acid

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Three dimensional flower-like NiO-thermally reduced Graphene Oxide (trGO) nanocomposite was prepared by a simple hydrothermal method. The morphology and structure of the nanocomposite was characterized using Scanning Electron Microscope (SEM), Transmission Electron Microscope (TEM), Energy Dispersive X-ray (EDX), X-ray Diffraction (XRD) and Fourier Transform Infrared (FTIR) spectroscopy. The electrocatalytic oxidation of Dopamine, Ascorbic Acid (AA) and Uric Acid (UA) was investigated using Cyclic Voltammetry (CV). The modified electrode NiO-trGO/GCE showed good electrocatalytic activity compared to bare GCE. The detection of dopamine was carried out using Differential Pulse Voltammetry (DPV) wherefrom the Limit of Detection (LOD) and linear range was estimated to be 50 nM and 10 μM to 500 μM respectively. The selective detection of dopamine in presence of UA and AA was carried out. The NiO-trGO/GCE showed excellent selectivity for the detection of dopamine in presence of UA. However AA was found to interfere in the determination of dopamine. Also, the validity of the present sensor was examined in human biological samples.

Keywords: NiO-trGO, Dopamine, Uric Acid, Ascorbic Acid, Electrochemical Sensor.

1. INTRODUCTION

Dopamine, belongs to the catecholamine family, is an important neurotransmitter in human Central Nervous System (CNS). It is secreted in the *substantia nigra* region of the human body and is responsible for muscular controls.[1] Abnormal dopamine levels leads to neurological disorders such as Parkinsons disease, Schizoprenia etc. The concentration of dopamine in the extracellular fluids of the Central Nervous System (CNS) is very low in the range of 0.01 μM–1 μM. Moreover, dopamine coexists with Ascorbic Acid (AA), Uric Acid (UA), adrenaline, noradrenaline etc. Among them, AA and UA are considered to be the common interfering agents in dopamine detection as they have a very close oxidation potential to that of dopamine.[2] Hence it is essential to detect dopamine selectively in presence of AA and UA. Diverse techniques were employed for the detection of dopamine such as chromatography,[3] fluorescence,[4] chemiluminescence,[5] electrophoresis,[6] electrochemical methods[7] etc. Among them, electrochemical method was widely accepted as simple, easy to operate and does not require laborious experimental conditions. High sensitivity in conjunction with selectivity can be achieved using electrochemical methods.

Nanoparticles with their small size and high surface area have widely attracted attention in diverse fields such as optics,[8] sensors,[9] photonics[10] etc. The properties of the nanomaterials can be fine tuned by changing the morphology and size of the nanomaterial in order to achieve different functionalities. In this context, metal nanoparticles of Au,[11,12] Ag[13] etc. have been studied towards electrochemical detection of dopamine. For instance, Au nanoparticles distributed on poly(4–aminophenol) have been investigated for selective detection of dopamine.[14] However metal nanoparticles lack stability as they can be leached out on subject to continuous potential cycling. On the other hand, metal oxides have good stability and high-corrosion resistance. Many Metal oxides such as ZnO,[15]

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SnO_2,[16] RuO_2,[17] Cu_2O,[18] CuO,[19] Co_3O_4,[20] TiO_2[21] have been employed for the detection of dopamine. Nickel oxide a *p*-type semiconductor has been studied as a potential material in various fields such as supercapacitors,[22–24] fuel cells,[25] magnetism,[26] and sensors.[27] The presence of multiple valence state of Ni makes them as an effective supercapacitor material. Impressively, NiO has also been used in detection of various molecules such as glucose,[28, 29] nitrite[30] etc.

Graphene, comprised of sp^2 hybridized carbon atoms packed in a honey-comb lattice like structure and has a high thermal and electrical conductivity as well as good mechanical strength.[31] Graphene has large theoretical surface area of 2600 m^2g^{-1}. However, pure graphene is hydrophobic and cannot be used without any modification. On the other hand, Graphene Oxide (GO) prepared from oxidation of graphite has large number of oxygen/carboxyl functional groups which renders high dispersion in solvents. The reduction of GO result in reduced Graphene Oxide (rGO); resembles more or less similar to graphene. The presence of residual oxygen/carboxyl functional groups, on the base and edge plane of rGO render dispersibility in solvents and were responsible for enhanced catalytic activity.[32] The rGO can be prepared from GO by thermal,[33] chemical,[34] and electrochemical[35] reduction methods. Generally it is known that rGO prepared by chemical reduction method exhibit good activity. Recently, Sivasubramanian et al[36] showed that chemically reduced GO exhibit high electrocatalytic activity towards dopamine detection.

As mentioned above shape controlled nanoparticles have gained increasing attention in recent years due to its easily tunable properties. Metal oxide nanostructures combined with graphene composite will impart synergistic effect leading to enhanced performance. In this paper, we report a facile synthesis of three dimensional hierarchical NiO-thermally reduced Graphene Oxide (trGO) nanocomposite for selective detection of dopamine in presence of AA and UA. The electrocatalytic activity was studied using Cyclic Voltammetry (CV) and the detection was carried out using Differential Pulse Voltammetry (DPV). The performance of the sensor in human biological samples was analyzed.

2. EXPERIMENTAL DETAILS

2.1. Chemicals

$NiCl_2 \cdot 6H_2O$ (99% Merck), Urea (99% Himedia), KH_2PO_4 (Merck, ≥98%), K_2HPO_4 (Merck, ≥98%), Dopamine hydrochloride (Himedia, ≥98%), AA (Lobachem, ≥99%), UA (Himedia, ≥99%) were employed without any purification process. Millipore water (resistivity 18 MΩcm) was used for the preparation of all the solutions.

2.2. Synthesis of NiO-trGO Nanocomposite

5 mg of graphite oxide powder prepared from modified Hummers method[37] was added to the pre-dissolved solution of 3.3 g of $NiCl_2 \cdot 6H_2O$ and 1.3 g of urea in 20 mL of water. The aqueous solution was transferred to the Teflon lined autoclave and heated at 150 °C for 3 hours. The resulting precipitate was centrifuged at 3000 rpm followed by water wash to remove any impurities. The precipitate was heated at 100 °C in oven for 8 h followed by calcinations at 400 °C for 2 h.

2.3. Material Characterization

The morphology of the nanocomposite was characterized using Scanning Electron Microscope (SEM) and Transmission Electron Microscope (TEM). The SEM image was obtained from FEI Quanta FEG 200 and the TEM image was recorded from JEOL JEM 2100. The elemental presence was confirmed using Energy Dispersive X-ray (EDX) analysis. The crystal structure was elucidated using X-ray Diffraction (XRD) pattern using Bruker D8 Advance equipped with copper Kα radiation and the bonding characteristics were studied using Fourier Transform Infrared (FTIR) spectroscopy. For TEM analysis, a few milligram of the sample was dispersed in ethanol followed by ultrasonication for 5 minutes, a small amount of the dispersed sample was dropcasted on carbon coated copper grids.

2.4. Electrochemical Investigations

CHI600E (CH Instruments, USA) with a conventional three electrode assembly was used to carry out all the electrochemical measurements such as CV, DPV and Electrochemical Impedance Spectroscopy (EIS). The Glassy Carbon Electrode (GCE) of 3 mm diameter was used as the working electrode and Platinum wire and Ag/AgCl were employed as counter and reference electrode respectively. All the electrodes were purchased from CH Instruments USA. The GCE was cleaned using various grades of alumina (0.3 micron and 1 micron) followed by ultrasonication. 1 mg of NiO-trGO nanocomposite was dispersed in 5 mL ethanol and 10 μL of the dispersed sample was dropcasted on the clean GCE surface. This NiO-trGO/GCE constitutes the working electrode and was employed for further studies. The EIS was carried out for bare GCE and NiO-trGO/GCE at their Open Circuit Potential (OCP) with 10 mV amplitude from a solution of 0.1 M PBS and 10 mM $K_4[Fe(CN)_6]$. The electrochemical response for dopamine, AA and UA was studied using CV on NiO-trGO/GCE from a freshly prepared 0.1 mM dopamine, 0.5 mM AA and 0.5 mM UA in 0.1 M Phosphate Buffer Saline (PBS) at pH 7 at a scan rate of 0.1 Vs^{-1}. The potential was scanned from −0.2 V to 0.8 V. The sensing was performed using DPV from 0.1 M PBS. The pulse width (0.05 s), pulse amplitude (0.05 V) and pulse period (0.5 s) were fixed to perform the DPV measurement. The DPV curves were recorded in the potential region between −0.2 to 0.5 V (vs. Ag/AgCl) after successive addition of various aliquots of dopamine, AA and UA.

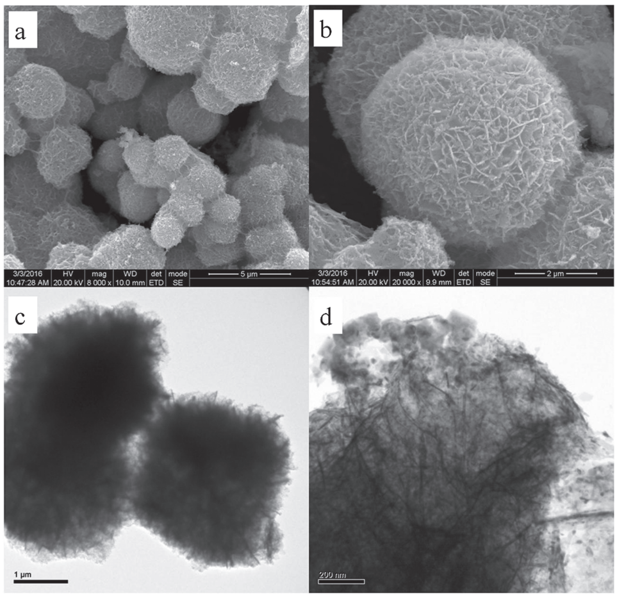


Figure 1. (a and b) depicts the SEM image of NiO-trGO nanocomposite prepared by a simple hydrothermal method and (c and d) shows the TEM image of the nanocomposite. The inset in 1d illustrates the corresponding SAED pattern.

2.5. Preparation of Real Samples

The estimation of dopamine and UA was carried out in human blood and urine samples on NiO-trGO/GCE. 100 μL of blood and urine sample was diluted 50 times using 0.1 M PBS solution. Known concentrations of dopamine and UA were spiked and the amount was estimated from the calibration plot obtained earlier.

3. RESULTS AND DISCUSSION

3.1. SEM and TEM Analysis

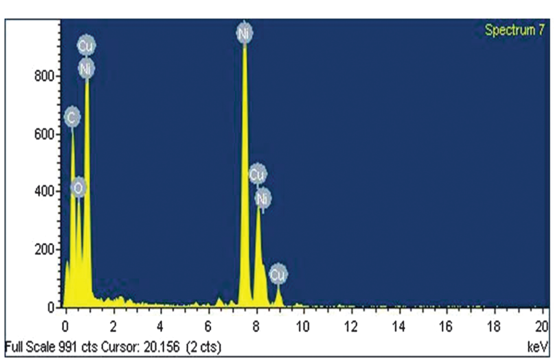
The morphology of the prepared nanocomposite was studied using SEM and TEM analysis. Figures 1(a and b) shows the SEM image of one-pot synthesized NiO-trGO nanocomposite obtained at different magnification. Three dimensional hierarchical flower-like structures were observed wherein the NiO and trGO were intertwined with each other, with the average size of the sphere ~10 μm. The corresponding TEM image of the nanocomposite was provided in Figures 1(c and d). The silhouette image of NiO-trGO nanocomposite was shown in Figure 1(c). Spherical shape with a diameter of ~7 to 8 μm was observed. Figure 1(d) depicts the high magnification TEM image of NiO-trGO nanocomposite, more number of wrinkles resembling the graphitic nature was noticed. The SAED pattern shown in the inset of Figure 1(d) exhibits a spot pattern indicating the crystalline nature of the nanocomposite. The EDX analysis was employed to confirm the presence of Ni, O and C elements and the corresponding weight percentage was found to be 62.0%, 4.66% and 8.38% respectively as shown in Figure 2.

3.2. XRD Analysis

Figure 3(a) depicts the XRD pattern of NiO, NiO-trGO nanocomposite, sharp diffraction peaks indicates the crystalline nature of the nanocomposite. In the case of NiO the diffraction peaks obtained at 37.3°, 43.2°, 62.9° and 75.5° corresponds to the (111), (200), (220) and (311) planes respectively. These values are consistent with JCPDS reference no. 22-1189 confirming the presence of Face Centered Cubic (FCC) structure of NiO. The diffraction pattern remains unchanged for the nanocomposite with the additional peak at 24.8° which corresponds to (002) plane of graphitic carbon atom. The (200) plane of NiO is of high intensity compared to other diffraction peaks obtained. No additional or secondary peaks was noticed which further confirms the high purity of the prepared nanocomposite.

3.3. FTIR Studies

The FTIR spectrum of NiO and NiO-trGO was depicted in Figure 3(b). The broad peaks at ~3500 cm^{-1} in both the spectrum corresponds to the OH stretching vibration of the adsorbed water molecule. For NiO, the peak at 1638 cm^{-1} was due to the OH bending vibration of water molecule. The peaks obtained at around 500 cm^{-1} represents the bending and stretching vibrations of Ni–OH and Ni–O respectively. However these peaks were slightly shifted in the NiO-trGO nanocomposite. The peak at 1741 cm^{-1}



Element	Weight%	Atomic%
C K	8.38	28.62
O K	4.66	11.95
Ni K	62.08	43.37
Cu K	24.88	16.06
Totals	100.00	

Figure 2. EDX image of NiO-trGO nanocomposite.

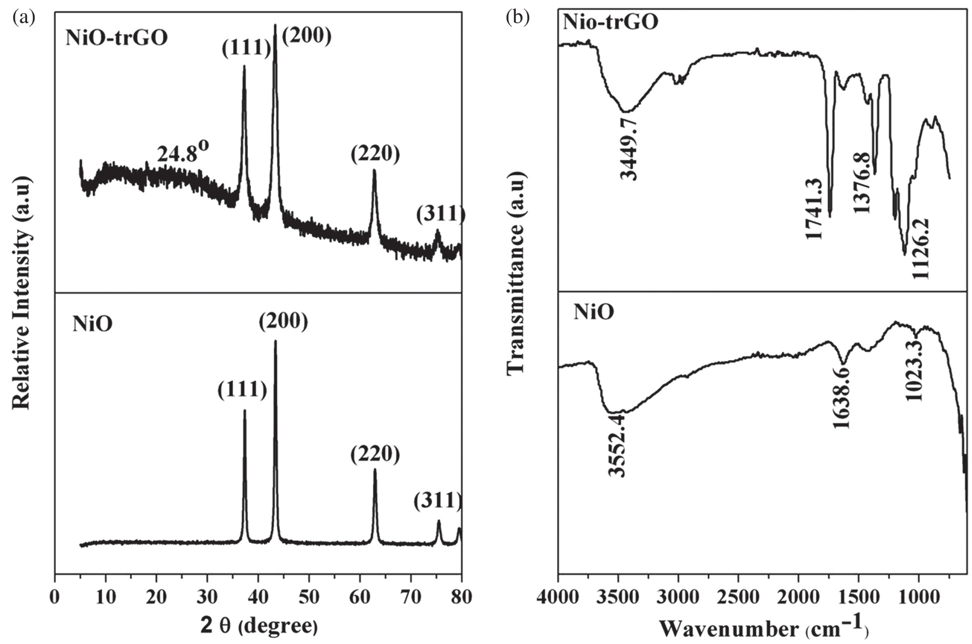


Figure 3. (a) XRD pattern and (b) FTIR spectra of NiO and NiO-trGO nanocomposite.

arise due to the C═O stretching vibration indicating the presence of residual COOH/CO functional groups in the composite. In addition the peak at 1368 cm^{-1} is for characteristic vibration of oxygen containing functional groups. These evidences suggest the existence of strong interaction between NiO and trGO in the nanocomposite. These results were in agreement with the NiO-rGO composite reported earlier by Jha et al.[38]

3.4. EIS Analysis

EIS is an effective tool to study the interfacial properties of the modified electrodes. Figure 4 shows the Nyquist plot of bare GCE and NiO-trGO/GCE recorded in the frequency range of 10^5 Hz to 1 Hz at their OCP from a solution of 0.1 M PBS and 10 mM $K_4[Fe(CN)_6]$. The Nyquist plots exhibit a typical semi-circle plot followed by a vertical spike corresponding to Warburg impedance due to diffusion. The diameter of the semicircle gives the charge transfer resistance (R_{ct}) which was found to be 1171 Ω and 632 Ω for bare GCE and NiO-trGO/GCE. The low R_{ct} of NiO-trGO/GCE indicates the excellent conductivity and electron transfer ability compared to bare GCE. The corresponding equivalent circuit has been provided in the inset of Figure 4.

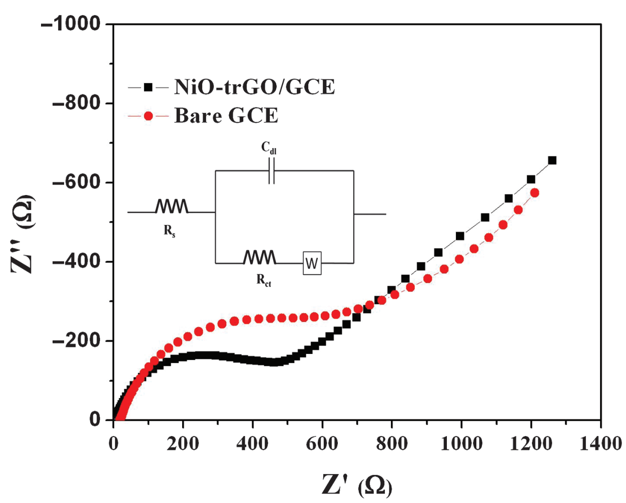


Figure 4. Nyquist plot for bare GCE and NiO-trGO/GCE from 0.1 M PBS and 10 mM $K_4[Fe(CN)_6]$ in the frequency range of 1×10^5 Hz to 1 Hz.

3.5. Electrocatalytic Activity Behavior of Dopamine, UA and AA on NiO-trGO/GCE

Figure 5(a) shows the CV curves for 0.1 M PBS solution on bare GCE and NiO-trGO/GCE at a scan rate of 0.1 Vs^{-1}. The NiO-trGO/GCE showed high current compared to bare GCE which may be attributed to increased surface area of the modified electrode. Figure 5(b) depicts the CV response for 0.1 mM dopamine in 0.1 M PBS on bare GCE and NiO-trGO/GCE. The oxidation peak for dopamine on bare GCE was obtained at 0.42 V with a peak current of 1.95 μA and the reduction peak at 0.01 V with peak current of -1.03 μA. On the other hand, for NiO-trGO/GCE the oxidation peak was noticed at 0.36 V with peak current of 3.89 μA and reduction peak at 0.03 V with peak current of -2.96 μA. A two fold increase in oxidation current was observed for NiO-trGO/GCE compared to bare GCE. In Figure 5(c), the CV curves represent the oxidation of 0.5 mM AA on both the electrodes. The oxidation peak was obtained at 0.63 V for bare GCE and 0.65 V for NiO-trGO/GCE. By comparing Figures 5(b and c), a

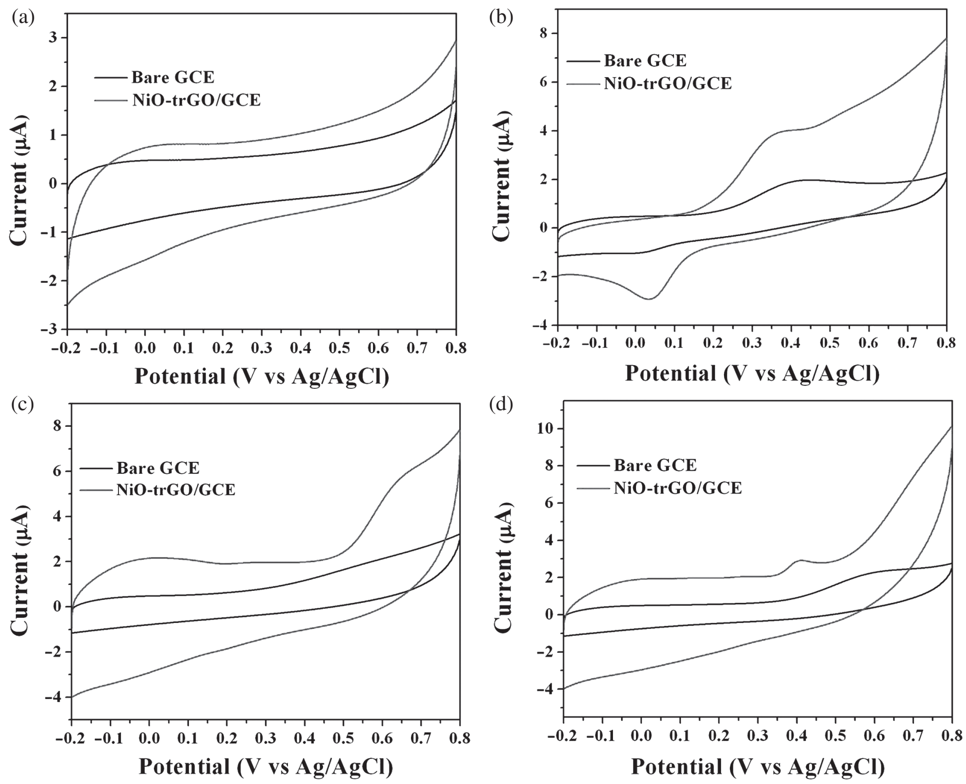


Figure 5. Cyclic voltammogramm using bare GCE and NiO-trGO/GCE in (a) 0.1 M PBS, (b) 0.1 mM of dopamine in PBS, (c) 0.5 mM of AA in PBS and (d) 0.5 mM of UA in PBS. Scan rate=0.1 V s^{-1}.

possible interference of AA in the detection of dopamine can be expected. In Figure 5(d) the CV curves represent the electrooxidation of UA from a solution of 0.5 mM UA and 0.1 M PBS. For bare GCE the oxidation peak potential was obtained at 0.58 V with current of 2.20 μA and the NiO-trGO/GCE showed the peak potential at 0.40 V with current of 2.91 μA. From the CV studies it can be inferred that the performance of NiO-trGO/GCE was better compared to bare GCE, however there could be a possible interference by AA in dopamine detection which was further studied using DPV. The influence of scan rate on the oxidation of dopamine was demonstrated. Figure 6 shows the voltammetric response for dopamine on NiO-trGO/GCE at different scan rate from 100 mV s^{-1} to 1000 mV s^{-1}. The peak current increases with increase in scan rate and a satisfactory linearity was obtained from the plot between i_p versus ν (Fig. 6(b)). The corresponding linear regression equation is given as follows $I_{pc} = -10.44\ \nu$ (mV s^{-1}) $-$ 3.49 ($R^2 = 0.9725$) and $I_{pa} = 8.54\ \nu$ (mV s^{-1}) + 3.25 ($R^2 = 0.9871$).

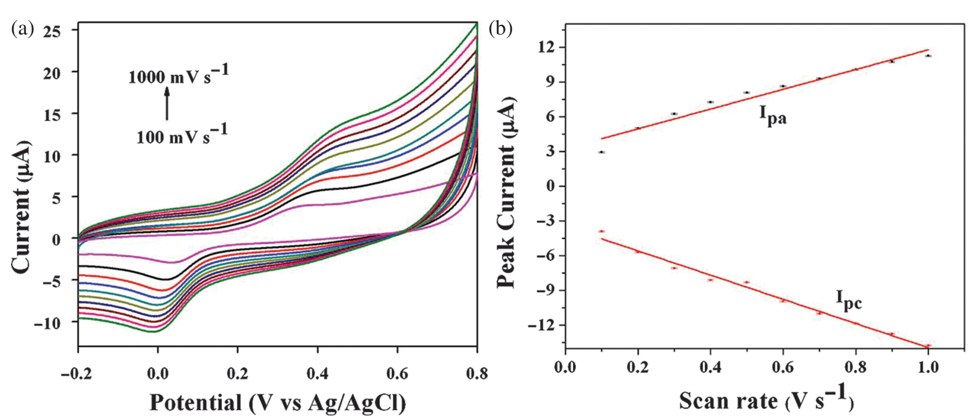


Figure 6. (a) CV response for dopamine on NiO-trGO/GCE from 0.1 M PBS and 0.1 mM dopamine solution obtained at different scan rates ranging from 100 mV s^{-1} to 1000 mV s^{-1} and (b) depicts the corresponding calibration plot.

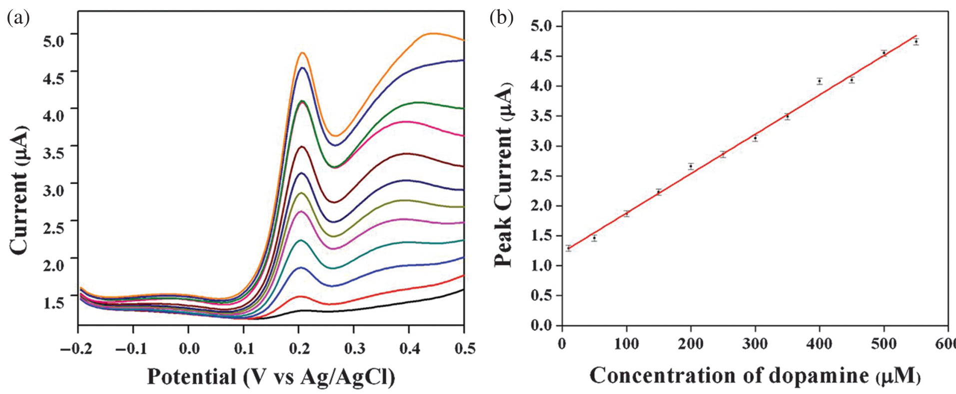


Figure 7. DPV curves for the sensing of dopamine using NiO-trGO/GCE from a solution of 0.1 M PBS at different concentration of dopamine ranging from 10 μM to 550 μM and (b) the corresponding calibration plot.

3.6. Sensing of Dopamine

The sensing of dopamine was carried out using DPV which is a highly sensitive technique for detection of analytes in trace levels. The sensing was carried out using NiO-trGO/GCE from a solution of 0.1 M PBS. A 10 mM stock solution of dopamine was prepared and various aliquots of dopamine concentration were added, subsequently the DPV curves were recorded by scanning the potential from -0.2 V to 0.5 V. Figure 7 shows the DPV response for addition of dopamine from the concentration range 10 μM to 550 μM and Figure 7(b) depicts the corresponding calibration plot. The peak current increases with the concentration of dopamine. The Limit of Detection (LOD) was estimated to be 50 nM with the signal to noise ratio as 3 and the linear range was obtained from 10 μM to 500 μM.

3.7. Selective Detection of Dopamine in Presence of UA and AA

The selective detection of dopamine in presence of UA and AA using NiO-trGO/GCE was performed. Figure 8 shows the DPV curves for varying concentration of dopamine in presence of 100 μM of UA. Two well-defined peaks corresponding to dopamine (at 0.25 V) and UA (at 0.35 V) were obtained. The dopamine-UA peak separation was found to be 100 mV which is suitable for selectively sensing dopamine in presence of UA. The peak current increases with increase in concentration. Figure 8(b) shows the calibration plot for different concentration of dopamine ranging from 50 μM to 350 μM, satisfactory linearity was obtained with the regression equation given as $I_p = 0.007C_{dopamine} + 2.37$ ($R^2 = 0.9761$). Similar studies have been carried out for UA in presence of 100 μM dopamine (Fig. 9). The DPV peaks show a linear variation with increasing concentration of UA from 100 μM to 500 μM and the regression equation is given as $I_p = 0.01C_{UA} + 2.27$ ($R^2 = 0.9961$). On the other hand, for AA the DPV studies were performed in presence of 200 μM of dopamine and 200 μM UA as shown in Figure 10. Interestingly as the concentration of AA increases, the DPV curves show a broad peak for all the three analytes. This leads to difficulty in analyzing dopamine and UA in presence of excess concentration of AA. The reason for such unusual behavior is still under study. A comprehensive illustration on the performance of the present sensor with that of other modified systems is provided in Table I. The performance

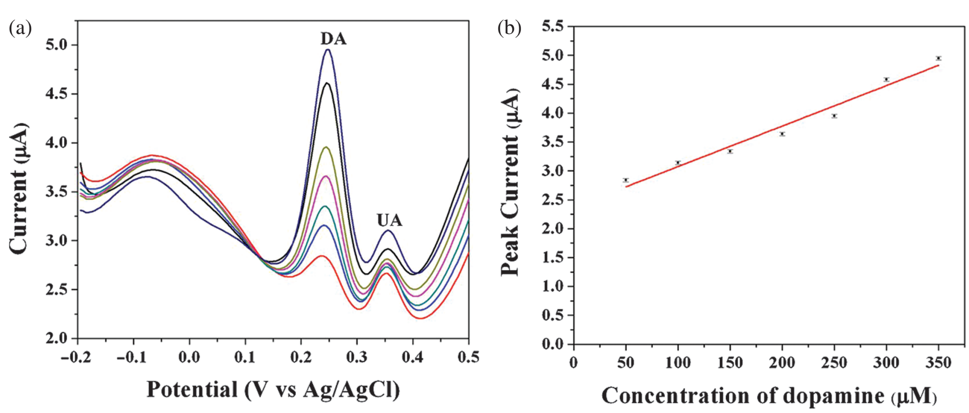


Figure 8. DPV curves for various concentration of dopamine from 50 μM to 350 μM in presence of 100 μM of UA on NiO-trGO/GCE and (b) the corresponding plot for concentration of dopamine with peak current.

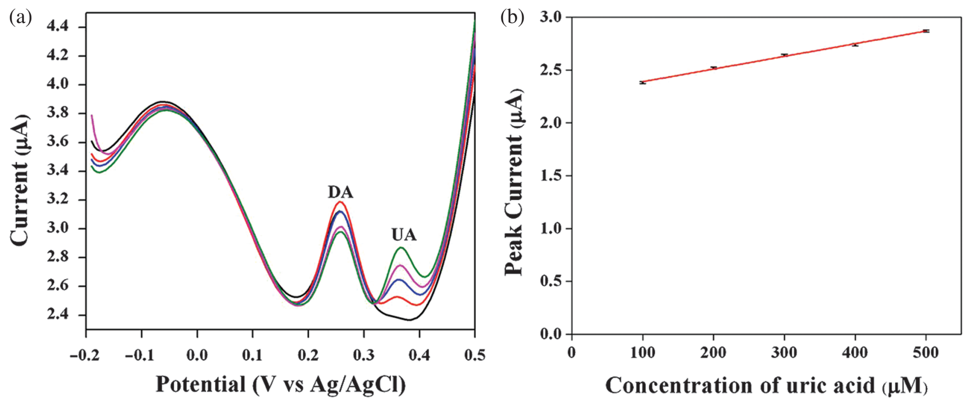


Figure 9. DPV curves for various concentration of UA from 100 μM to 500 μM in presence of 100 μM of dopamine on NiO-trGO/GCE and (b) the corresponding plot for concentration of uric acid with peak current.

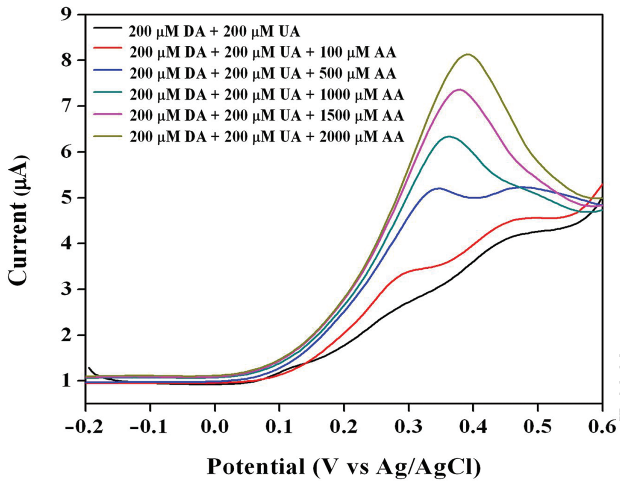


Figure 10. DPV curves for various concentration of AA in presence of 200 μM of dopamine and 200 μM of AA and 0.1 M PBS on NiO-trGO/GCE.

of the modified electrode was superior compared to Ag NPs/rGO[40] and Fe_3O_4@Au[47] nanoparticles in terms of detection limit. Also, no obvious changes was observed in the DPV peaks of dopamine in the presence of ten-fold excess concentration of Na^+, K^+, Cl^-, NO_3^- and urea solution.

3.8. Repeatability and Stability

The repeatability of NiO-trGO/GCE for the detection of dopamine was investigated from a solution of 100 μM dopamine. The peak current was measured for ten successive runs and the results were satisfactory with least significant standard deviation of 1.3%. This indicates the excellent repeatability of the sensor. The long term stability of the NiO-trGO/GCE modified electrode was studied by measuring the peak current of 100 μM dopamine successively for 10 days. The electrode was stored in PBS solution after measurement. It was found that the peak current retained 87% of the original value for dopamine.

3.9. Unknown Concentration Determination and Analysis of Real Sample

The validity of the proposed NiO-trGO/GCE electrode towards dopamine detection was studied. Various concentration of dopamine was analyzed and the amount of dopamine was estimated by comparing the current obtained for each DPV measurement with the calibration plot obtained in Figure 7(b). Each experiment was repeated

Table I. A comparison on the performance of different electrodes for sensing of dopamine, UA and AA.

			LOD (μM)			Linear range (μM)			Peak separation (mV)			
S. N	Material	Method	DA	UA	AA	DA	UA	AA	UA-DA	DA-AA	AA-UA	Ref.
1.	trGO-TiO_2	DPV	–	–	–	1–1000	1–900	10–1000	300	100	450	[39]
2.	AgNP/rGO	DPV	5.4	8.4	9.6	10–800	10–800	10–800	219	140	359	[40]
3.	Au ME	DPV	–	–	–	1-100	–	–	266	124	384	[41]
4.	PPD/MWCNT	DPV	0.5	0.5	5	10–1000	1–100	5–1200	248	232	480	[42]
5.	C-Pt/TiO_2	DPV	–	–	–	1–10	10–60	–	180	160	320	[43]
6.	AuNP/Ty/GO	DPV	0.05	–	–	0.5–411	–	–	15	140	155	[44]
7.	Au NP/overoxidized polyindole	DPV	–	–	–	5–68	6–486	210–1010	124	128	668	[45]
8.	EACPE/CPE	DPV	0.08	0.1	6	6–100	6–90	400–1000	320	485	136	[46]
9.	Fe_3O_4@Au	DPV	0.1	0.2	1	0.5–50	1–90	6–350	240	130	410	[47]
10.	NiO-trGO	DPV	0.05	10	100	10–500	100–500	–	100	–	–	This work

Table II. Prediction of unknown concentration of dopamine using NiO-trGO/GCE.

Amount of dopamine added (μM)	Amount of dopamine estimated (μM)	% of recovery
100	106.8	106.8 ± 2
150	162.3	108.2 ± 3
250	259.1	103.6 ± 2
300	348.6	99.6 ± 4
450	503.1	111.8 ± 2

Table III. Determination of dopamine and UA in real samples.

	Amount added (μM)		Amount estimated (μM)			
Sample	Dopamine	UA	Dopamine	% of recovery	UA	% of recovery
Blood	100	–	88.7	88.7 ± 2	–	–
	150	–	143.8	95.8 ± 3	–	–
	175	–	171.6	98.0 ± 2	–	–
Urine	50	100	51.4	102.7 ± 2	108.7	108.7 ± 3
	100	250	93.1	93.1 ± 4	244.4	97.7 ± 3
	150	500	138.1	92.0 ± 5	466.6	93.3 ± 4

thrice and Table II indicates the amount of dopamine taken and the amount estimated after the analysis. The values are satisfactory with least significant standard deviation and excellent recovery values. The recovery values were calculated using the formula: % of recovery $= C_{\text{predicted}}/C_{\text{added}} \times 100$.

Human blood and urine samples were employed to test the performance of the sensor by spiking a known amount of dopamine and UA to the solution. Prior to addition, the samples were diluted 50 times with 0.1 M PBS. Table III provides the estimated values of dopamine and UA, the values were found to be satisfactory with least significant standard deviation.

4. CONCLUSION

A simple hydrothermal method was employed for the synthesis of NiO-trGO nanocomposite. The morphological analysis using SEM and TEM studies reveal the presence of three-dimensional flower-like NiO-trGO nanocomposite with average diameter of 10 μm. The XRD pattern reveals the presence of Face Centered Cubic structure and the structural analysis was carried out using FTIR spectroscopy. The electro-oxidation of dopamine, UA and AA was studied using CV wherefrom enhanced catalytic activity was observed for NiO-trGO/GCE compared to bare GCE. A limit of detection of 50 nM with a linear range from 10 μM to 500 μM for dopamine was estimated from DPV. The selective detection of dopamine was investigated using DPV in presence of AA and UA. The proposed sensor showed excellent selectivity towards dopamine detection in presence of UA. However, a high concentration of AA seems to influence the detection of dopamine. The validity of the proposed sensor was examined in human biological samples.

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